DAY: ______ DATE: ______

Sample estimate,

Sample mean of population Parameter Mean (μ):

The sample mean:

$$\bar{X} = \frac{\Sigma X}{n}$$

is a sample estimator (unbaised) of the population parameter mean (μ)

Example:

2, 5, 7

$$\bar{X} = \frac{2+5+7}{3} = \frac{14}{3} = 4.67.$$

Sample estimation of population Parameter Variance (σ^2) and S.D σ:

$$\text{Diff of squares} = \frac{\Sigma X^2 - \left(\frac{(\Sigma X)^2}{N}\right)}{N}$$

mean of squares / Square of means.

DAY:

DATE:

Sample Variance (S^2)

$$S^2 = \frac{\Sigma (X - \bar{X})^2}{n}$$

is the biased (or standard) estimator of population variance. (σ^2)

Population

$$\text{Variance} = \sigma^2 = \frac{\Sigma (X - \bar{X})^2}{N-1}$$

Sample Standard Deviation (S).

$$S = \sqrt{\frac{\Sigma (X - \bar{X})^2}{n-1}} = \sqrt{\frac{\Sigma X^2 - \frac{(\Sigma X)^2}{n}}{n-1}}$$

Unbaised estimator of S.D (σ).

Represents the square root of variance.

Measure dispersion of data ($X - \bar{X}$).

Based on mean of squared deviations.

DAY: ______ DATE: ______

$$S^2 = \frac{1}{n-1}\left[\Sigma X^2 - \frac{(\Sigma X)^2}{n}\right]$$

S^2 is the estimated unbaised estimator of population variance (σ^2).